

Definition of Convergence in Probability

- $\frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty$.
- (ϵ, δ) -definition of limit: $\lim_{x \rightarrow c} f(x) = L$. The definition uses ϵ to represent any small positive number, so that "f(x) becomes arbitrarily close to L" means that f(x) eventually lies in the interval $(L - \epsilon, L + \epsilon)$, The phrase "as x approaches c" then indicates that we refer to values of x, whose distance from c is less than some positive number δ , that is, values of x within either $(c - \delta, c)$ or $(c, c + \delta)$,

- In this section, we formalize a way of saying that a sequence of random variables is getting "close" to another random variable.
- Let $\{X_n\}$ be a sequence of random variables and let X be a random variable defined on a sample space. We say X_n converges in probability to X if for all $\epsilon > 0$

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| \geq \epsilon) = 0,$$

or

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| < \epsilon) = 1.$$

If so, we write

$$X_n \xrightarrow{p} X.$$

- If $X_n \xrightarrow{p} X$, we often say that the mass of the difference $X_n - X$ is converging to 0.

- Chebyshev's Inequality:

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

- **Weak Law of Large Numbers:** Let $\{X_n\}$ be a sequence of iid random variables having common mean μ and variance $\sigma^2 < \infty$. Then,

$$\bar{X}_n \xrightarrow{P} \mu, \quad \text{as } n \rightarrow \infty.$$

- **Theorem 4.2.2.:** Suppose $X_n \xrightarrow{P} X$ and $Y_n \xrightarrow{P} Y$. Then,
 $X_n + Y_n \xrightarrow{P} X + Y$.
- **Theorem 4.2.3.:** Suppose $X_n \xrightarrow{P} X$ and a is a constant. Then $aX_n \xrightarrow{P} aX$.
- **Theorem 4.2.4.:** Suppose $X_n \xrightarrow{P} a$ and the real function g is continuous at a . Then $g(X_n) \xrightarrow{P} g(a)$.
- If $X_n \xrightarrow{P} a$, then $X_n^2 \xrightarrow{P} a^2$; $1/X_n \xrightarrow{P} 1/a$, if $a \neq 0$; $\sqrt{X_n} \xrightarrow{P} \sqrt{a}$ if $a \geq 0$.
- **Theorem 4.2.5.:** Suppose $X_n \xrightarrow{P} X$ and $Y_n \xrightarrow{P} Y$. Then $X_n Y_n \xrightarrow{P} XY$.

$$S^2 \xrightarrow{p} \sigma^2?$$

Let $X_1, \dots, X_n$ be iid from a distribution with mean μ and variance $\sigma^2 < \infty$.

$$\frac{1}{n} \sum_{i=1}^n X_i^3 \xrightarrow{p} ???$$